INSTITUTE OF DISTANCE LEARNING
KWAME NKRUMAH UNIVERSITY OF SCIENCE AND TECHNOLOGY, KUMASI, GHANA

AE 464
CROP STORAGE AND STORAGE STRUCTURES

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Durable crop storage unit 2

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Durable Crop Storage 2 (Bag storage)

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CONVENTIONAL BAG STORAGE

- The most common method of grain storage in many countries is bag storage.
- It may be done on raised platforms, in ventilated cribs, protected movable covers or in stores or warehouses.

Bag Storage





CONVENTIONAL BAG STORAGE

- The Principal criteria required for proper storage in warehouses include:

  - Satisfactory control of humidity and temperature
  - Possibility of fumigation
  - Effective protection against rodents

Requirements and Functions of a Warehouse

- Keeps grain secure from deterioration by insect pests, rodents and birds, poor storage conditions/inclement weather and theft
- Efficiently receives grain from traders and dispatches grain to end-users
- Good and efficient management of grain inventory
- Easy maintenance of warehouse structure
- Satisfactory working conditions for staff and laborers

Planning for Grain Storage

- The following should be considered:

– Volume of grain handled through the facility
– Average Length of storage (short-term 3 to 6 months or long-term over 12 months)
– Storage for specific target market, strategic reserve program, transit facility or warehouse receipts program

Capacity of Warehouse

- Warehouses can be small (50-200 tonnes capacity), medium (500-2,000 tonnes capacity) or large (5,000-10,000 tonnes capacity).

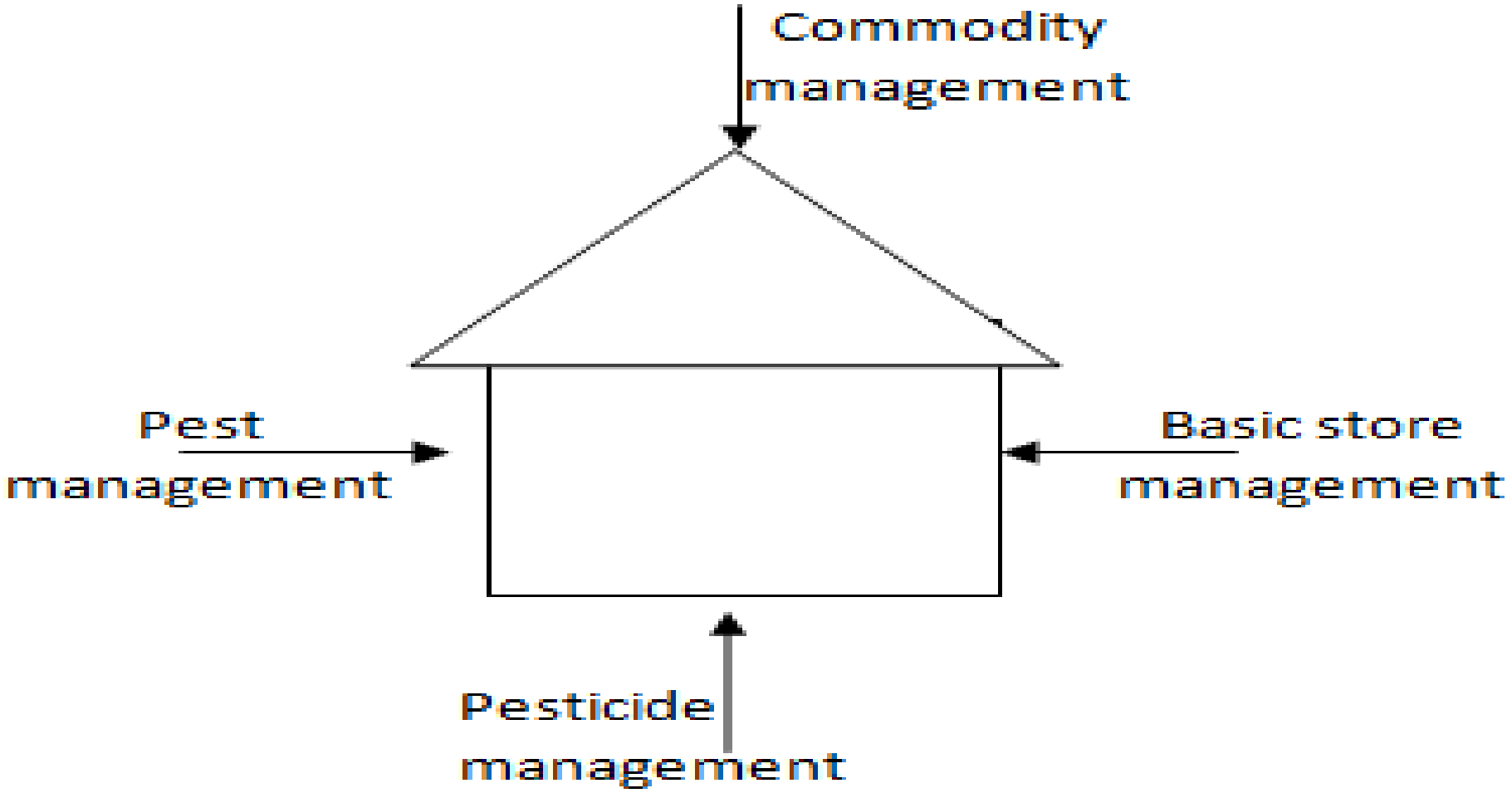
- For small warehouses, the usable volume can be less than 50% of the gross volume below the eaves level, and increases to 70-80% for 5,000 to 10,000 tonnes capacity

- For bagged commodity, the usable volume is less than the warehouse gross volume because:

  – All spaces above the eaves level should be left for air to circulate and ventilate

  – A clear space of at least 1 m must be left around roof supports, and between the stack and wall,

  – A clear space of at least 2 m must be left between stacks



Factors to consider in warehouse management

Factors considered in the design of warehouses for storage

- Siting and orientation
- Warehouse should be constructed on a raised site
- Good drainage
- Firm soil for construction
- Good road connection to the warehouse
- Wind direction: should be considered in the siting of the wall direction

General construction features

Floors

- Floors should be raised at least 1m above the ground in other to prevent ground water from entering the store.
- They should serve as a vapour or moisture barrier.
- Concrete floors should be hard enough to bear the expected loads – this is to prevent cracks.
- Smooth surface floors help in cleaning.

Walls

- Walls should have a smooth surface and should be without cracks
- Metal walls are usually not encouraged to be used.

Roofing

- The roof overhang should be at least 1m to ensure that the warehouse is kept in shade and its walls are protected against rain.

Roofing

- There should also be properly sealed connection between the roofs and the walls as this will prevent birds from entering.
- In some cases, roof gutters are necessary for water collection and also to protect the foundation.

Doors

- Metal doors are usually used. If wooden doors are used, the lower part should be of metal.

Warehouse Structure and Layout

- Commodity storage area
- Grain cleaning, drying and bagging station
- Grain quality analysis lab
- Receiving station (weigh bridge or large scale)
- Temporary storage shed
- Lorry (truck) parking and good traffic flow

Attributes of grain warehouse operations

- An efficient grain receiving & dispatch system
- Good quality control system
- Good inventory management
- Good management practices
- Good personnel

Good grain receiving & dispatch practice

- The receiving system is the first point of management of the receiving stock.
- Poorly managed receiving operations can lead to problems in managing the stored commodity.

A good receiving and dispatch system should have:

- – a thorough and systematic sampling plan of incoming grain
- – an efficient flow of incoming and outgoing grain such that piles of unprocessed grain are not created
- – good tracking of customers loads from scale to stacking or loading
- – good record keeping, labor management and security

Basic Tools in Quality Control Lab

- -Bag trier to sample grain
- -Moisture meter
- -Temperature probe
- -Sampling sieves
- -Aflatoxin Kit



Bag Trier/ Parkhi
Model MRC 708

Good Inventory management

- Know the grain standards you need to meet
- Practice "First In First Out", FIFO: this ensures that you maintain the newest stocks of grain in inventory.
- FIFO practice must be tailored based on the prevailing incoming quality of new grain.
- Practice good record keeping
- Tag inventory of grain with recorded quality

Keep track of inventory and management with floor maps

- Floor maps help identify the inventory location at the facility and keeps a map of problem spots which enable better management and decision making
- Indicate fumigated stacks
- Indicate premium quality stacks
- Indicate stacks that need turnover

Practice D.I.C.E

- - Dry grain before bagging
- - Inspect warehouse before storing grain
- - Clean warehouse and surroundings regularly
- - Examine grain regularly

Good personnel management

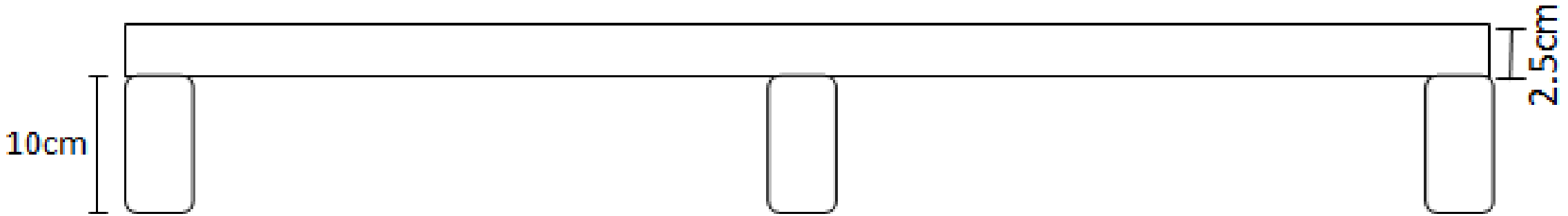
- Ensure operators and managers are trained and are aware of their responsibilities
- Facilitate recurrent training of operators, managers and their staff
- Recognize shortcoming of operations staff /staff needs
- Bring all personnel on board about facility operations
- Develop rapport between grain suppliers and warehouse management staff
- Develop a reward system to incentivize staff to be better operators and managers

Stacking of bags

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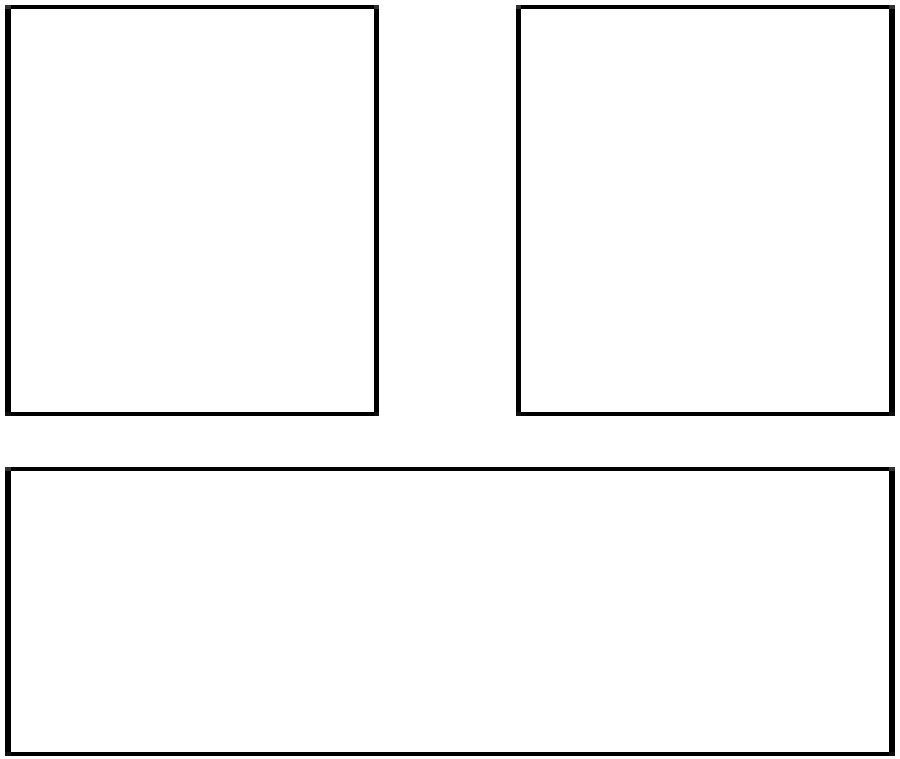
Stacking of bags

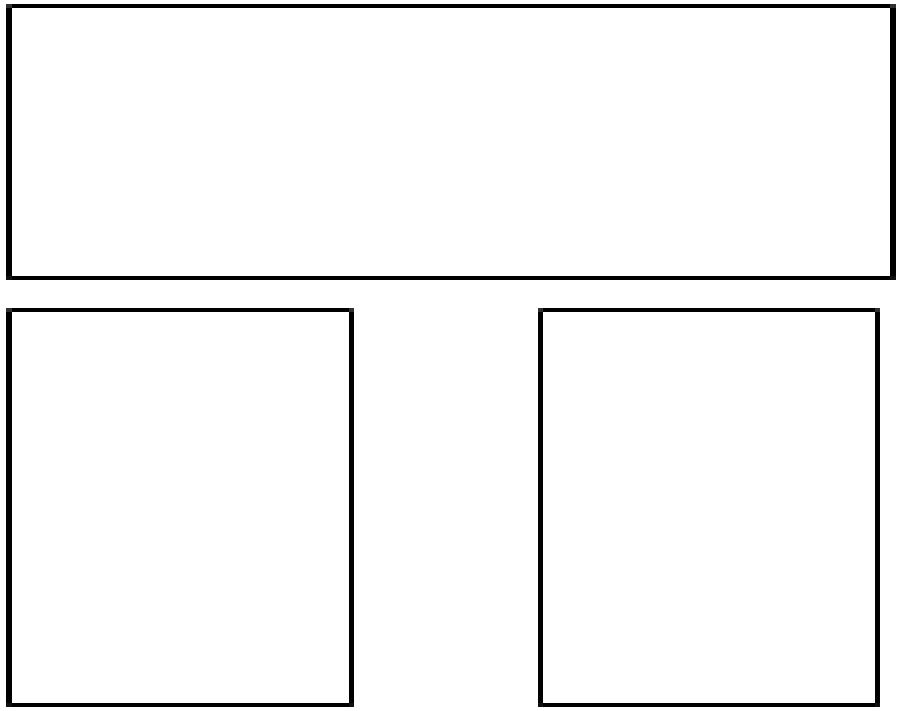
- It is done on pallets in range from 3, 5 and 8. Sacks being used for storage should be stacked in a way that allows air circulation.
- The pallets are usually kept 10cm off the ground.



Packing of the bags is done from one end to the other end.

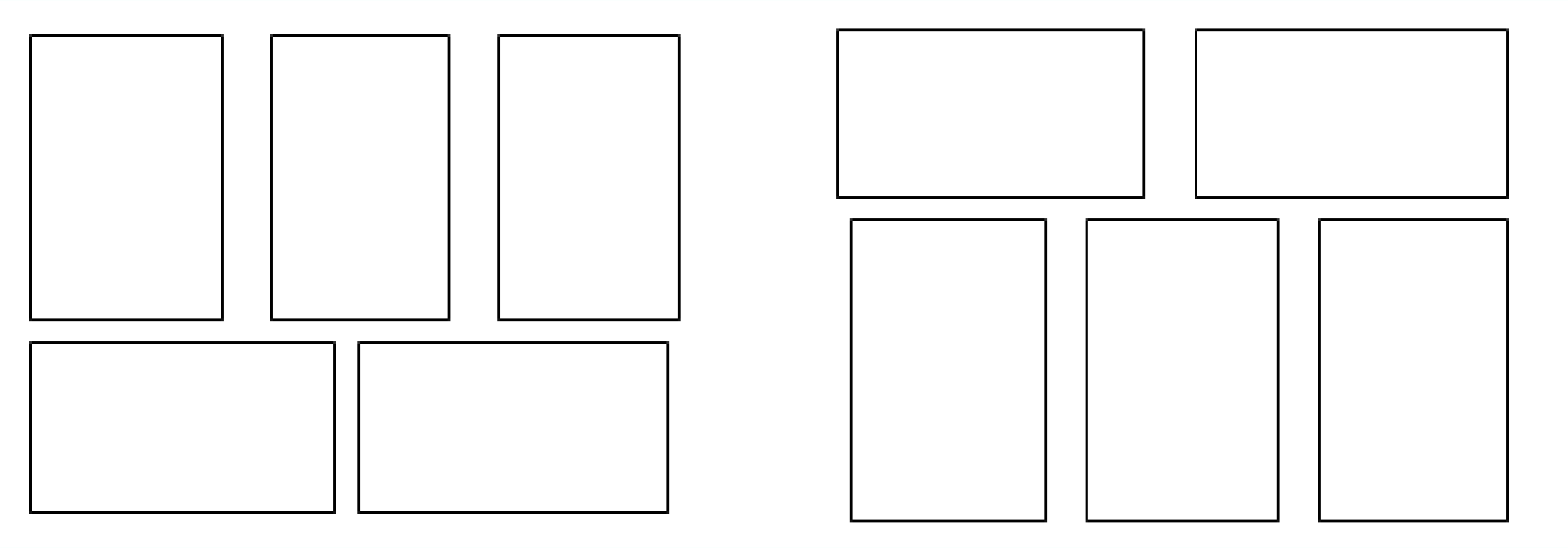
3 Pack Arrangements





Alternating bags placed lengthwise / widthwise

5 Pack Arrangements



Alternating bags placed lengthwise / widthwise

- Space is provided between the wall and the stack and within the stacks
- The bags are placed on pallets
- The layers of bags are arranged perpendicularly and in a regular fashion
- The store is cleaned
- A space of at least 1m must be left between the top of a bag-stack and the roof for operations

STORE CAPACITY

- In calculating the capacity of a store the volume to be occupied by a given tonnage of grain is determined.
- The following information is required in the determination of the storage volume:
  - Type of grain
  - Useful surface area
  - Stowage factor volume (m^3/kg)
  - Bulk density (kg/m^3) or specific

- **Stowage factor** is the reciprocal of bulk density.
- Units: m^3 /tonne

Food stuff	Stowage factor (m^3 /tonne)
Beans (bulk)	1.3
Millet (bags)	1.2
Maize (bulk)	1.4
Maize (bags)	1.6
Maize (heads or cobs)	2.2
Groundnut (seed in bags)	1.3
Groundnut (in shells, bulk)	
Rice (paddy in bags)	2.0

Steps for store capacity determination

- Determine the surface area
- Determine the useful surface area: Surface area- space occupied by gangways
- Determine useful volume by multiplying the useful surface area by the height of wall or expected height of product
- Determine the tonnage

Example

- Consider a rectangular store 60m long by 12m wide and 3m high. There are two 1m wide gangways running lengthwise along the walls and a central gangway 4m wide running lengthwise. Also running widthwise are two 1m wide gangways along the walls plus another three 4m wide gangways. (Stowage factor 1.6, bag is 100kg)

- Assume bagged maize are stacked to the top of the walls determine the maximum number of bags of maize to be stored in this warehouse. Assume height of stack to be 3m

Solution

- Useful length: 60 -1-1- 4 – 4 - 4 = 46m
- Useful width: 12- 1- 1- 4 = 6m
- Useful surface area = 46m x 6m =276m^2
- Useful volume = 276m^2 x 3m = 828m^3
- Tonnage = Useful volume/stowage factor
- = 828m^3 / 1.6m^3/tonne = 517.5 tonnes
- Number of bags = 517.5 x 1000kg/ 100kg =5175bags
- **Note:** The extent to which the available volume is used increase with the size of building.

Utilisation Factor (UF)

- It is the proportion of floor usage in a warehouse. It is the ratio of the floor area covered with grain to the total floor area of the store.
- **UF (%) = FA_g/FA_t**
- **FA_g(**floor area covered with grain) =276m^2
- **FA_t (**total floor area) =720m^2
- **UF =** 38.33%

MANAGEMENT OF WAREHOUSES

- Maintenance of equipment and warehouses
- Correct handling and storage of the products
- Management of the stored products
- Regular control of the storage facilities
- Keep correct records
- Write reports regularly
- Control of insects and rodents – treat building, repair cracks in walls, clean building
- Bag stacking should be properly done to maximise space and maintain hygiene
- Keep surrounding of warehouse clean

6. VENTILATION/AERATION IN STORAGE

- Moisture movement is linked to the deterioration of products in storage (moulding).

- When grains are kept in storage heat and moisture are generated. These are undesirable as it will enhance the rate of deterioration of the product.

- Relative humidity of the air has to be checked in order not to exceed equilibrium relative humidity of the given moisture content. Air is therefore blown through stored grain (referred to as **Aeration**) to reduce moisture within he store.

6.2 AERATION IN WARE HOUSES/ BAG STORAGE

- The system usually used is refrigeration type system or dehumidification system.

- Cooling Unit: fans, evaporation, condensation

Thank You

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